Homework 3

EN.601.439/639 | Fall 2026

You must submit either a physical handwritten homework or print out your homework to submit at the beginning of class.

Problem 1: Graph-regularized NMF

Suppose we are given (1) a single gene expression matrix $X \in \mathbb{R}^{N \times G}$ which measures the expression of G genes in N cells and (2) a graph $\mathcal{G} = (V, E)$ called a *biological interaction network* whose vertices $V = \{1, \ldots, G\}$ are the observed genes and where an edge $(g, g') \in E$ connects genes g and g' that interact in some way (e.g. their proteins interact or the genes are co-expressed).

We would like to factorize the single cell data matrix as $X \approx UV$, where $U \in \mathbb{R}^{N \times K}$ is the factor matrix and $V \in \mathbb{R}^{K \times G}$ is the loading matrix, such that the loadings are "smooth" on the graph $\mathcal{G}$; i.e., if there is an edge $(g, g') \in E$ in the graph $\mathcal{G}$, then the corresponding columns $V_{\cdot,g}$ and $V_{\cdot,g'}$ in the loading matrix are close.

Write an NMF-style optimization problem for computing such a factorization. (Hint: use the standard NMF optimization problem and add an additional term to the objective function which encourages the loadings to be "smooth").

Problem 2: Deriving special cases of multidimensional scaling (MDS)

We are given the distance $d_{ij} > 0$ between data points $i, j \in \{1, \ldots, N\}$ for N datapoints. Suppose we are told there exist unknown vectors $\mathbf{z}_1, \ldots, \mathbf{z}_N \in \mathbb{R}^K$ such that

$$d_{ij} = ||\mathbf{z}_i - \mathbf{z}_j||_2 \tag{1}$$

for all pairs $i, j \in \{1, \ldots, N\}$ of datapoints.

In this question, we will derive an explicit algorithm for computing the vectors $\mathbf{z}_i$.

a) Prove that $-\frac{1}{2} J D_2 J = G$, where

- $J = I - \frac{1}{N}\mathbf{1}^T\mathbf{1} \in \mathbb{R}^{N \times N}$, where $I \in \mathbb{R}^{N \times N}$ is the identity matrix and $\mathbf{1} \in \mathbb{R}^{N \times 1}$ is an all-ones vector;
- $D_2 = [(d_{ij})^2] \in \mathbb{R}^{N \times N}$ is a matrix of squared distances; and
- $G = [g_{ij}] \in \mathbb{R}^{N \times N}$ has entries $g_{ij} = \mathbf{z}_i^T \mathbf{z}_j$, and is called the *Gram matrix*.

b) Let $Z = [\mathbf{z}_1 | \cdots | \mathbf{z}_N] \in \mathbb{R}^{N \times N}$ be a matrix whose columns are the vectors $\mathbf{z}_1, \ldots, \mathbf{z}_N$. We call Z the *embedding matrix*. Prove that $G = Z^T Z$.

c) Let $G = Q\Lambda Q^T$ be the eigendecomposition of the Gram matrix G. Derive an equation expressing the unknown embedding matrix Z in terms of the matrices Q, Λ. [Note: there are multiple possible answers.]

d) Suppose you are given the squared-distance matrix D_2. Give an algorithm for computing the embedding matrix Z.

e) Suppose the distances d_{ij} are *Euclidean distances*; that is, $d_{ij} = ||\mathbf{x}_i - \mathbf{x}_j||_2^2$, where $\mathbf{x}_1, \ldots, \mathbf{x}_N \in \mathbb{R}^G$ are the data vectors. Show that $-\frac{1}{2}JD_2J = XX^T$, where $X \in \mathbb{R}^{N \times G}$ is the data matrix with rows $\mathbf{x}_1, \ldots, \mathbf{x}_N$. In this special case, what is your algorithm in part d equivalent to? (Hint: it is one of the first methods we studied in the course.)

Problem 3: Random walk variants

Let G be a connected, unweighted graph with adjacency matrix A and degree matrix D.

a) Define a lazy random walk with parameter $\delta \in (0, 1)$ as follows. For a walker at vertex v, with probability δ the walker stays at vertex v and with probability $1 - \delta$ the walker moves to an adjacent node, selected uniformly at random.

Derive an equation for the probability transition matrix P and the stationary distribution π for the lazy random walk. Furthermore, show that the lazy random walk has period 1, so that the random walk always converges to the stationary distribution.

b) Define a **random walk with restart** with restart parameter $r \in (0, 1)$ and a fixed node v as follows. At each step of the random walk, with probability $1 - r$ the walker moves to an adjacent node, and with probability r the walker "teleports" to v.

Derive an equation for the probability transition matrix P and the stationary distribution π for the random walk with restart.

c) Express the stationary distribution π from part b in terms of an infinite sum by using the identity $(I - A)^{-1} = \sum_{k=1}^{\infty} A^k$ (for sufficiently nice matrices A). Intuitively, what does each term in the infinite sum mean?

Problem 4: Diffusion Distance

Let $P \in \mathbb{R}^{N \times N}$ be the transition matrix of a random walk on a graph. Assume that P is a symmetric matrix and has eigendecomposition $P = V\Lambda V^T$ where $V = [v_1|\cdots|v_N] \in \mathbb{R}^{N \times N}$ is a matrix of eigenvectors and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_N)$ is a diagonal matrix of eigenvalues.

Define the diffusion distance between vertices x and y as

$$D_t(x, y) = \|P^t_{\cdot,x} - P^t_{\cdot,y}\|, \tag{2}$$

where $P^t_{\cdot,x}$ is the x-th column of the exponentiated transition matrix P^t and describes the distribution of a random walker that starts at vertex x after t steps.

Prove that the diffusion distance $D_t(x, y)$ has the following closed-form expression:

$$D_t(x,y) = \sqrt{\sum_{i=1}^{N} \lambda_i^{2t} \big(v_k(x) - v_k(y)\big)^2}. \tag{3}$$

Problem 5: Conditional Probabilities in t-SNE

We are given data points $x_1, \ldots, x_N \in \mathbb{R}^K$. Recall that for each pair x_i, x_j of datapoints, t-SNE defines the quantity

$$p_{j|i} = \frac{\exp\left(-\|x_i - x_j\|^2/(2\sigma_i^2)\right)}{\sum_{k \neq i} \exp\left(-\|x_i - x_k\|^2/(2\sigma_i^2)\right)} \tag{4}$$

which is called a *conditional probability.*

a) Suppose point x_j is far from point x_i, i.e. $\|x_i - x_j\| \gg \sigma_i$, while at least some other points remain close to x_i. Show that $p_{j|i} \approx 0$. What does this tell us about the neighborhood of datapoint i that t-SNE looks at?

b) If $\|x_i - x_j\| \ll \sigma_i$, show that $p_{j|i} \approx \dfrac{1 - \frac{\|x_i - x_j\|^2}{2\sigma_i^2}}{\sum_{k \neq i}\left(1 - \frac{\|x_i - x_k\|^2}{2\sigma_i^2}\right)}$. *[Hint: use the approximation $e^{-a} \approx 1 - a$ for small a.]*

Show why, in this case, t-SNE is similar to multidimensional scaling (MDS).